



University of Sri Jayewardenepura

Faculty of Computing

**ACADEMIC WORKSHOPS PARTICIPATION
& ENGAGEMENT ANALYSIS**

CIS3062 - Business Intelligence

Group Number – Group 6

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1. Introduction

1.1 Background

Technology-related academic workshops play a vital part in improving technical knowledge, skills and industry exposure of students. Student organizations in the University of Sri Jayewardenepura constantly hold workshops on programming, artificial intelligence, cyber security, web development, cloud computing, software engineering and other emerging technologies. These workshops allow students with opportunities for hands-on experience that cannot be obtained in the classroom.

Organizers record the registration information and feedback from each workshop in a separate spreadsheet, in a registration form, and in a feedback form/questionnaire, etc. This makes it challenging to study participation patterns, to benchmark workshop effectiveness, and to assess the level of student engagement.

Business Intelligence (BI) is a useful solution to make unstructured data structured and presentable in the form of dashboards and visualizations. Organizers can use the data collected from the BI to monitor key metrics, analyze participation patterns, evaluate the effectiveness of the workshops, and inform future planning efforts.

This project aims to create a 'Business Intelligence' dashboard for analysis of participation and engagement in technology-related academic workshops conducted by the various student groups and organizations in the University by employing Microsoft Power BI.

1.2 Importance of Business Intelligence

Business Intelligence allows organizations to convert raw data into useful information that can aid in better decision-making.

In this project, BI will help organizers to:

- Evaluate the involvement in several workshops.
- Determine the most popular categories of technical workshops.
- Be aware of participation over time.
- Compare rates of participation by faculty and academic year.
- Analyze participant feedback.

- Provide insights to support future workshop planning with data.

1.3 Real-World Relevance

Each year there are various academic workshops organised by the student organizations. However, future workshops are decided on experience rather than on data analysis.

The aim of the project is to illustrate the application of Business Intelligence in a real University environment, to analyse the participation and engagement in the workshops. The dashboard can help organizers choose appropriate workshop topics, refine promotion efforts, boost student engagement, and make future events better.

2. Problem Statement

Academic workshops based on technology are held regularly to enhance Technical knowledge and professional skills of the students. The collected information, such as registration forms, attendance data, and feedback from participants is typically stored in separate systems without the ability to be analyzed in a centralized way.

This makes it hard for organizers to know what kinds of workshops to expect to see what works best, how well the workshops are being carried out, what kinds of workshops to expect to see, and how engaged students are at the workshops.

If the analysis is not done, future workshop planning will be less effective, since most of the decisions will be made on a basis of assumption.

Thus, a Business Intelligence dashboard integrating data from the various student organizations, and providing meaningful information to support data-driven decision making is needed.

3. Objectives

3.1 Main Objective

To develop a Business Intelligence dashboard that analyzes participation and engagement in technology-related academic workshops organized by student organizations.

3.2 Specific Objectives

- To compare student involvement in various workshops.
- To determine which workshop categories are the most popular in terms of technology.
- To compare the participation across faculties.
- To analyze participation based on the academic year.
- To find the trend in participation over time.
- To recap the comments of the participants.
- To make suggestions for better future workshops.

4. Methodology

4.1 Data Collection Method

The data used in this study was collected from registration forms and participant feedback forms of technology-related academic workshops organized by various student organizations at the University of Sri Jayewardenepura. The selected workshops included Algo Ace, Beauty of Cloud, Glow Up, Skill Up, TechXplore 2.0, Google Women Techmakers, and Algo Arena. The workshops took place in various technical areas such as programming, cloud computing, software engineering, cybersecurity, mobile application development, UI/UX, career development, and entrepreneurship.

The registration and feedback records were obtained in Microsoft Excel format after receiving permission from the respective student organizations. All personally identifiable information, such as participant names, email addresses, phone numbers, registration numbers, IEEE membership numbers and other sensitive information, were removed prior to use of the data for analysis to maintain participant privacy and confidentiality.

The rest of the data was compiled into a standardised dataset that included the workshop information, participant demographics and responses to the questions. The Business Intelligence analysis was performed with the help of Microsoft Power BI, with this standardized dataset being the main source of data.

4.2 Dataset Description

The final dataset combines registration and feedback information collected from multiple technology-related academic workshops organized by different student organizations. As different workshop registration and feedback formats were used in the original workshops, so the data that were collected were converted into a common format for analysis. The standardized dataset is a combination of three primary tables containing workshop details, participant registration information, and participant feedback, together with several supporting dimension tables used for data modeling in Power BI.

The dataset includes information related to workshop details, participant universities, faculties, academic years, workshop categories, registration records, and participant feedback. These attributes allowed the analysis of workshop participation patterns, participant demographics, workshop performance, and overall engagement.

Dataset Summary

Item	Description
Number of workshops analyzed	6
Number of organizers	4
Total registrations	1080
Total feedback responses	212
Study period	2025 year

Dataset Attributes

Attribute	Description
Registration ID	Unique identifier for each participant
Workshop ID	Unique identifier for each workshop
Workshop Series	Name of the workshop
Organizer	Student organization conducting the workshop
Category	Subject area of the workshop
Date	Date of the workshop
University	Participant's university
Faculty	Participant's faculty
Academic Year	Participant's academic year
USJ Student	Whether the participant is from the University of Sri Jayewardenepura
Satisfaction Rating	Participant satisfaction score
Speaker Rating	Rating given by participants to the workshop speaker
Content Relevance	Participant's rating of how relevant the workshop content is
Suggestions	Participant recommendations for improving future workshops
Feedback	Participant comments

4.3 Tools Used

Several software tools were used throughout the development of the project.

- **Google Forms** served as the original platform used by student organizations to collect workshop registration details and participant feedback for several workshops included in this study.
- **Microsoft Excel** was used to collect, integrate, clean, preprocess, and standardize data obtained from different workshop registration and feedback forms.
- **Microsoft Power BI Desktop** was used to perform data modeling, create relationships between datasets, develop DAX measures, and build interactive dashboards for data analysis and visualization.

4.4 Data Cleaning and Preparation

The collected registration and feedback datasets were received from different student organizations in various formats. The data was inconsistent in terms of column names, missing values and data structures in each workshop due to the use of different registration forms and feedback questionnaires. For this reason, some preprocessing was performed to prepare the data to be standardized for analysis.

First, all personally identifiable information, including participant names, email addresses, phone numbers, registration numbers, and IEEE membership numbers, was removed to maintain privacy and confidentiality.

Next, the column names from different workshop datasets were standardized into a common format. Similar attributes, such as university, faculty, academic year, workshop details, and feedback responses, were mapped to consistent column names to allow all workshop data to be combined into a single dataset.

Categorical values were also standardized. Faculty names, academic year values, workshop categories, and Yes/No responses were converted into consistent formats to eliminate duplicate representations caused by spelling differences or inconsistent naming. Duplicate records were identified and removed where necessary, while missing values were retained as blank where the information was unavailable.

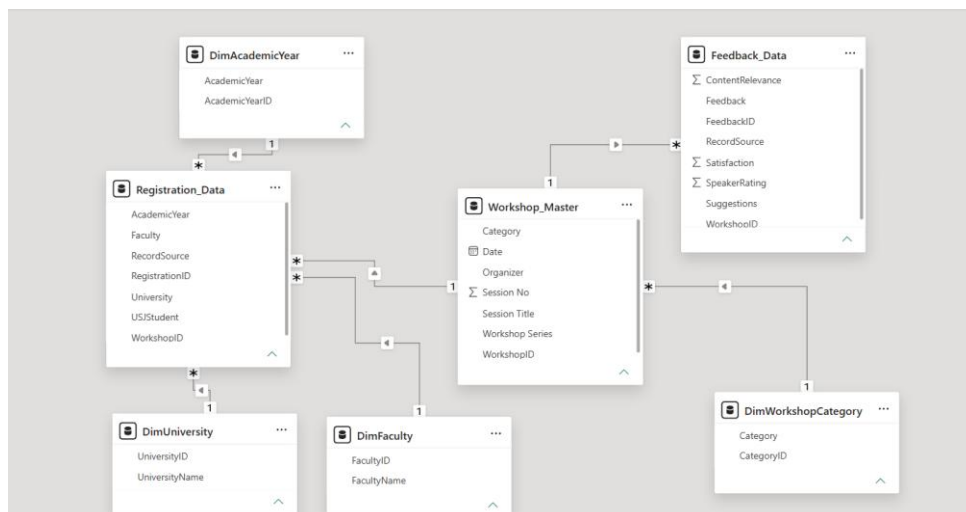
To improve data organization and support efficient analysis, the dataset was divided into three primary tables: **Workshop_Master**, **Registration_Data**, and **Feedback_Data**. Additional dimension tables, including **DimUniversity**, **DimFaculty**, **DimAcademicYear**, and **DimWorkshopCategory**, were created to normalize the data and reduce redundancy.

Also each workshop session was assigned a unique **WorkshopID**, which served as the primary key for linking workshop information with participant registrations and feedback records. This standardized structure provided a clean and reliable dataset that was ready for data modeling and analysis in Microsoft Power BI.

4.5 Data Modeling

After the data preparation process, the cleaned datasets were imported into Microsoft Power BI Desktop for data modeling. A star schema was implemented to organize the data efficiently and improve report performance.

The **Workshop_Master** table was used as the central dimension containing workshop information such as Workshop ID, Workshop Series, Session Title, Organizer, Category, and Date. Two fact tables: **Registration_Data** and **Feedback_Data** stored participant registration records and feedback responses respectively. Also supporting dimension tables, including **DimUniversity**, **DimFaculty**, **DimAcademicYear**, and **DimWorkshopCategory**, were connected to the fact tables through one-to-many relationships.



The relationships were established using the **WorkshopID** field together with the relevant dimension attributes. So this data model reduced data redundancy, improved data consistency, and enabled efficient filtering across multiple dashboard pages.

In addition to the relationships, several DAX measures were created to calculate key performance indicators, including Total Registrations, Total Workshops, Total Feedback Responses, Average Satisfaction, Feedback Response Rate, and participant counts. These measures formed the analytical foundation of the interactive Business Intelligence dashboard.

4.6 Dashboard Design and Visualization Approach

An interactive dashboard was developed using Microsoft Power BI to offer meaningful insights into student participation and engagement in technology-related academic workshops. The dashboard was developed to support data-driven decision-making by allowing users to filter and explore information from different perspectives through interactive visualizations and slicers.

The dashboard consists of four main report pages:

- **Executive Overview** presents high-level key performance indicators (KPIs), including total workshops, total registrations, total feedback responses, average satisfaction, and overall participation trends.
- **Participant Analysis** examines participant demographics by faculty, university, academic year, and student category to understand the characteristics of workshop attendees.
- **Workshop Performance** compares workshop series, individual sessions, categories, and organizers to identify the highest and lowest performing workshops based on participant registrations.
- **Feedback Analysis** evaluates participant satisfaction, speaker ratings, content relevance, sentiment distribution, and participant suggestions to assess workshop quality and identify opportunities for improvement.

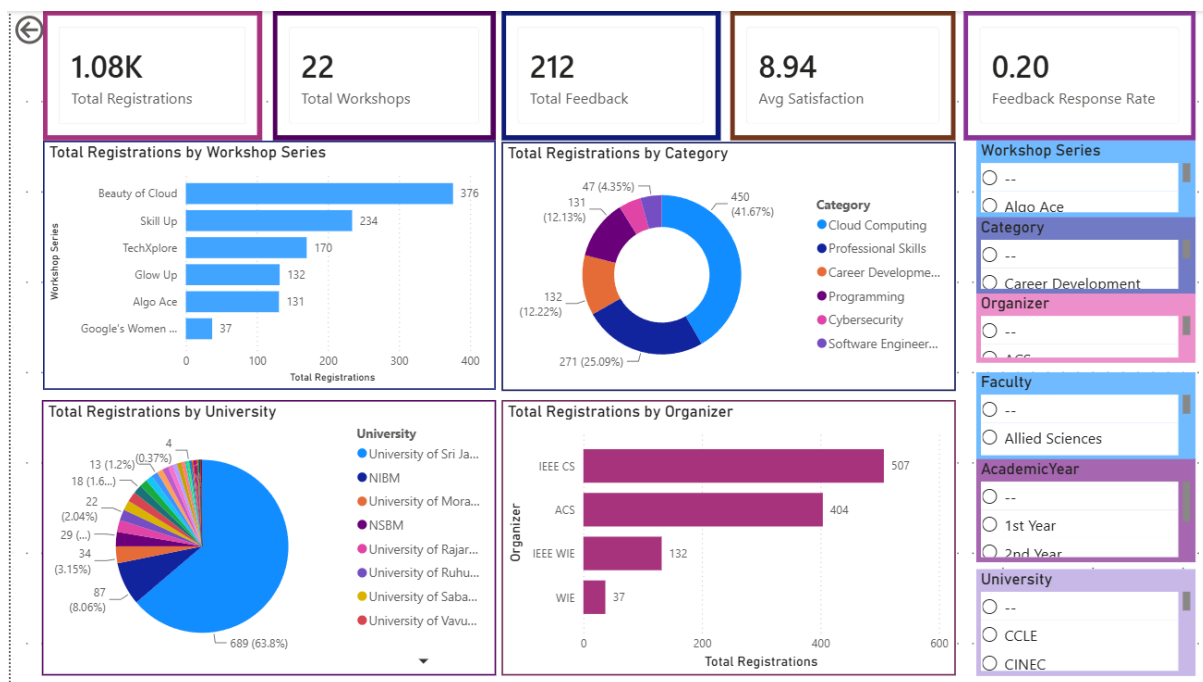
The dashboard incorporates a variety of interactive visualizations, including KPI cards, clustered bar charts, column charts, line charts, pie and donut charts, tables, and slicers. These visual elements enable users to explore participation patterns, compare workshop performance, and generate meaningful business insights through an intuitive and interactive interface.

5. Results and Discussion

The developed Business Intelligence dashboard is an interactive platform for analyzing participation and engagement in technology-related academic workshops conducted by various student organizations. The dashboard has four report pages, each focusing on a different part of the collected data. So through the use of interactive visualizations, users can explore workshop participation patterns, participant demographics, workshop performance, and participant feedback to support informed decision-making.

The following sections discuss each dashboard page and interpret the insights obtained from the visualizations.

5.1 Executive Overview



The Power BI dashboard provides a comprehensive overview of participant registrations, workshop performance, and participant feedback for the Academic Workshops Participation & Engagement Analysis project. The dashboard consolidates information from workshop registration and feedback datasets to support data-driven evaluation of workshop engagement and performance.

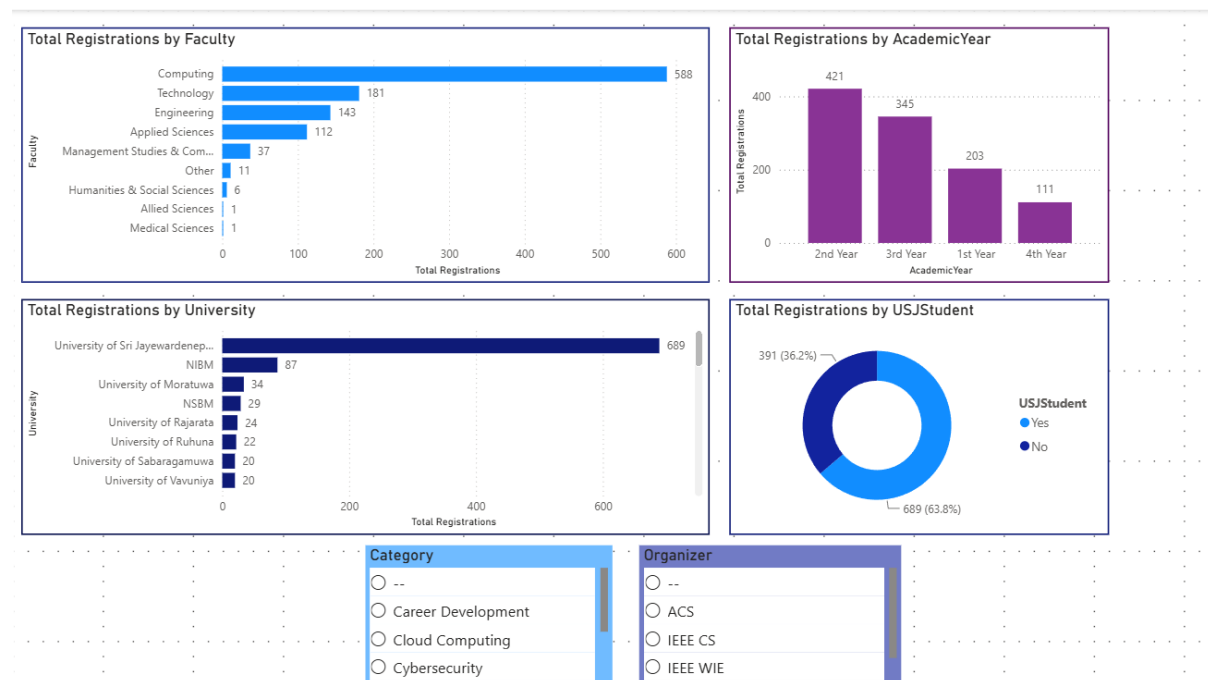
A total of **1,080 registrations** were recorded across **22 workshops**, with **212 feedback responses** collected from participants. The dashboard reports an **average satisfaction score of**

8.94 and a **feedback response rate of 0.20 (20%)**, indicating that approximately one in five registered participants completed the post-workshop feedback survey.

The workshops focused on various technical and professional development areas, including Cloud Computing, Programming, Artificial Intelligence, Cybersecurity, Professional Skills, Career Development, and Entrepreneurship. So the interactive slicers enable users to analyse registrations by workshop series, session title, organizer, category, faculty, academic year, and university, allowing stakeholders to explore participation patterns from multiple perspectives.

Overall, the dashboard offer valuable insights into participant demographics, workshop popularity, organizer performance, and participant satisfaction, supporting informed decision-making for future workshop planning.

5.2 Participant Analysis



The participant analysis highlights significant differences in registrations across workshop series, categories, universities, faculties, and academic years.

University participation was highly concentrated. The **University of Sri Jayewardenepura** accounted for **689 registrations (63.8%)**, significantly exceeding every other participating institution. **NIBM** ranked second with **87 registrations (8.06%)**, followed by the **University of Moratuwa**, **NSBM**, **University of Rajarata**, **University of Ruhuna**, **University of Sabaragamuwa**, **University of Vavuniya**, and **SLIIT**, each contributing comparatively

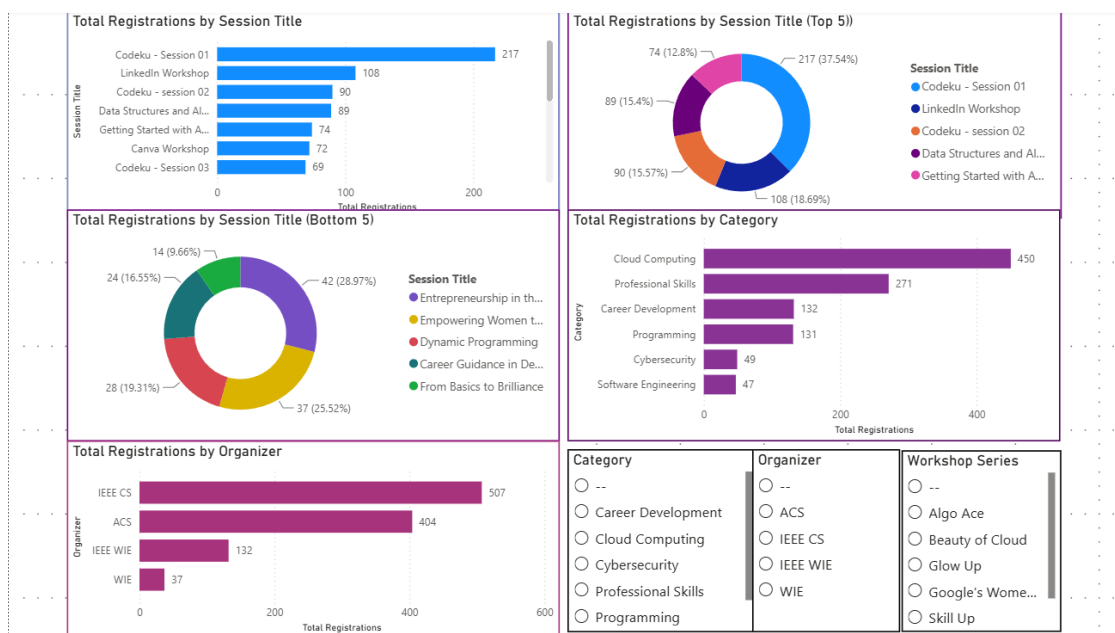
smaller participant numbers. Numerous additional universities contributed only a small fraction of registrations, demonstrating broad institutional representation despite the dominance of one university.

Faculty analysis revealed that the **Computing** faculty generated the largest number of registrations, with approximately **590 participants**. **Technology** ranked second with approximately **180 registrations**, followed by **Engineering** with approximately **145 registrations** and **Applied Sciences** with approximately **110 registrations**. Faculties such as Management Studies & Commerce, Humanities & Social Sciences, Allied Sciences, Medical Sciences, and Other contributed relatively fewer participants.

Academic year analysis indicates that **second-year students** represented the largest participant group with approximately **430 registrations**, followed by **third-year students** with approximately **340 registrations**. **First-year students** contributed approximately **200 registrations**, while **fourth-year students** recorded the lowest participation among the visible year groups.

The dashboard also illustrates the distribution of participants based on USJ student status. **689 participants (63.8%)** were identified as students from the **University of Sri Jayewardenepura**, while **391 participants (36.2%)** represented external institutions. This demonstrates that although the workshops primarily served students from the host university, they also attracted considerable participation from other higher education institutions.

5.3 Workshop Performance



The workshop performance analysis provides detailed insights into the popularity of individual workshop sessions, workshop categories, and organizing bodies.

Among the individual workshop sessions displayed, **Codeku – Session 01** recorded the highest participation with **217 registrations (37.54%)**, making it the most successful session. The **LinkedIn Workshop** ranked second with **108 registrations (18.69%)**, followed by **Codeku – Session 02** with **90 registrations (15.57%)**, **Data Structures and Algorithms** with **89 registrations (15.40%)**, and **Getting Started with AI** with **74 registrations (12.80%)**. Other sessions, including **Canva Workshop** and **Codeku – Session 03**, also attracted notable participant numbers, although lower than the leading sessions.

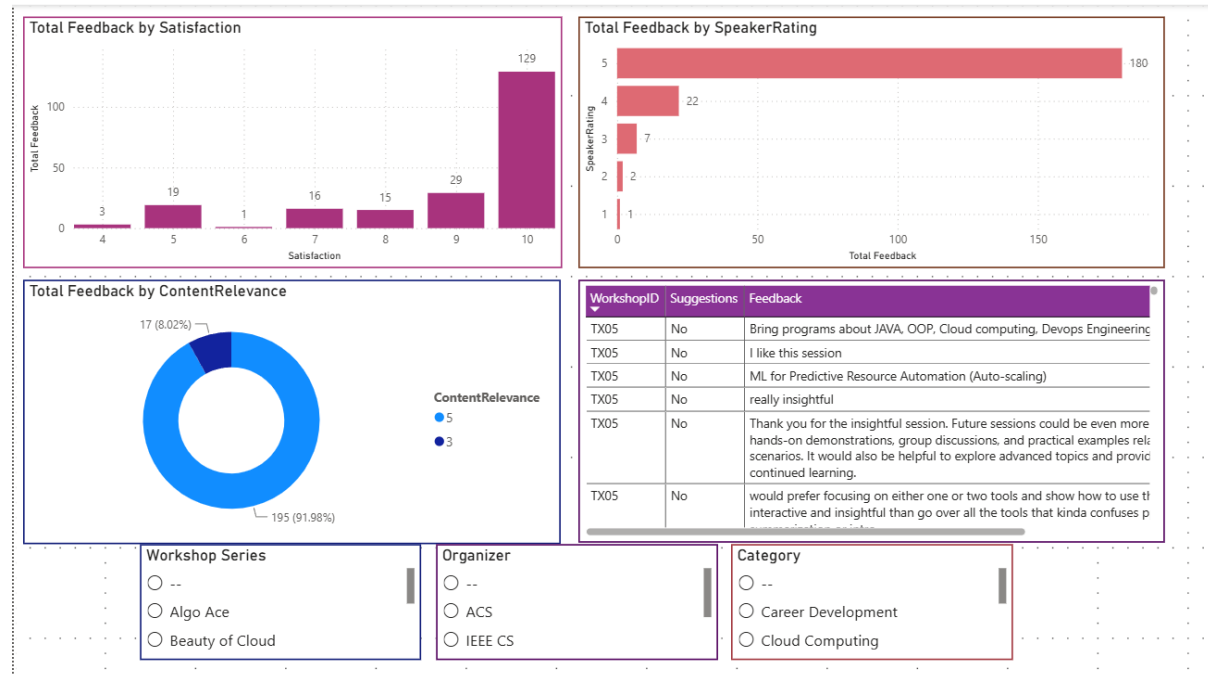
Another visualization highlights registrations across additional workshop sessions. **Entrepreneurship in the digital era** recorded **42 registrations (28.97%)**, followed by **Empowering Women in Tech** with **37 registrations (25.52%)**, **Dynamic Programming** with **28 registrations (19.31%)**, **Career Guidance in Development** with **24 registrations (16.55%)**, and **From Basics to Brilliance** with **14 registrations (9.66%)**. Compared with the larger workshop series, these sessions attracted more balanced participant numbers.

Category-wise analysis further confirms that **Cloud Computing** was the most popular workshop category with **450 registrations**, followed by **Professional Skills** with **271 registrations**. **Career Development** and **Programming** each attracted over **130 participants**, while **Cybersecurity** and **Software Engineering** recorded comparatively fewer registrations. These findings suggest that students demonstrated greater interest in cloud technologies and professional development than in more specialized technical domains.

The organizer analysis indicates that **IEEE CS** was the most successful organizing body, attracting approximately **500 registrations**. **ACS** ranked second with approximately **400 registrations**, while **IEEE WIE** attracted approximately **130 registrations**. The standalone **WIE** organizer recorded the smallest participant count. These results demonstrate the strong organizational reach and popularity of IEEE CS workshops throughout the programme.

Overall, the workshop performance analysis indicates that technical workshops focusing on cloud technologies and employability skills consistently generated the highest participant engagement, while organizer performance varied considerably across different workshop series.

5.4 Feedback Analysis



Participant feedback was analysed using satisfaction scores, speaker ratings, content relevance, and participant suggestions.

The dashboard indicates that **212 feedback responses** were collected from approximately **1,000 registrations**, resulting in a **feedback response rate of 20%**. Although the response rate is relatively modest, the feedback received provides valuable insights into participant perceptions of workshop quality.

The satisfaction distribution shows that the majority of respondents awarded the highest satisfaction rating of **10**, followed by ratings of **9**. Comparatively few participants selected ratings between **4** and **8**, indicating that overall satisfaction was consistently high across the workshops.

Speaker evaluations were similarly positive. Most respondents assigned a **speaker rating of 5**, while a smaller number selected **4**, and only a very limited number awarded ratings between **1** and **3**. This distribution suggests that workshop presenters effectively communicated the workshop content and maintained high presentation quality.

Content relevance received exceptionally favourable evaluations. Of the visible responses, **195 participants (91.98%)** rated the workshop content with the highest score of **5**, while **17 participants (8.02%)** assigned a rating of **3**. No other content relevance ratings are visible, indicating that participants overwhelmingly perceived the workshop material as useful and relevant to their academic or professional interests.

The dashboard also displays participant suggestions submitted for various workshops. Several visible records, including **TX05**, **BOC03**, **GWT01**, and **TX01**, contain "**No**" under the Suggestions field, indicating that many respondents chose not to provide additional written recommendations despite submitting numerical ratings. This pattern suggests that participants were generally satisfied with the workshops but were less inclined to provide qualitative feedback.

Overall, the feedback analysis demonstrates consistently positive participant perceptions regarding workshop satisfaction, speaker effectiveness, and content relevance. However, the relatively low feedback response rate and limited number of written suggestions indicate an opportunity to improve participant engagement during the post-event evaluation process. Encouraging more participants to submit detailed qualitative feedback would provide organizers with richer insights for future workshop improvements.

6. Key Business Insights and Recommendations

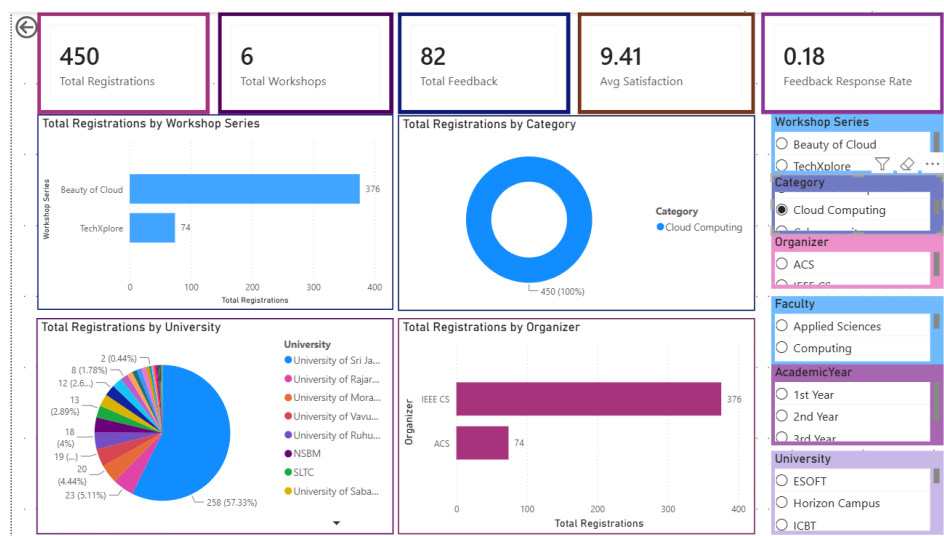
The interactive dashboard provides several insights into student participation and engagement in technology-related academic workshops. These findings can assist student organizations in evaluating previous workshops and making informed decisions when planning future academic events. The following insights were obtained from the dashboard analysis.

6.1 Key Business Insights

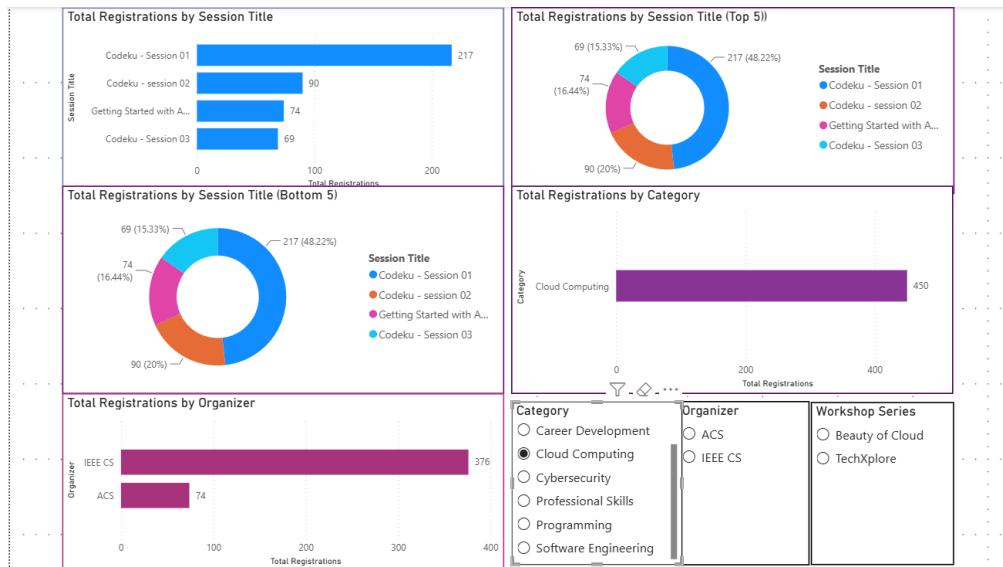
The Power BI dashboard reveals several important findings regarding participant engagement, workshop popularity, organizer performance, and participant satisfaction.

The analysis shows that approximately **1,000 participants** registered across **22 academic workshops**, demonstrating strong overall engagement with the workshop programme. However, only **212 participants** submitted feedback, resulting in a **20% feedback response rate**, indicating that while workshop attendance was high, participant engagement with post-event evaluation was comparatively low.

Workshop participation was highly concentrated in specific technical areas. **Cloud Computing** emerged as the most popular category with **450 registrations (41.67%)**, followed by **Professional Skills** with **271 registrations (25.09%)**. Career Development and Programming workshops also attracted substantial participation, whereas Cybersecurity and Software Engineering workshops received comparatively fewer registrations. These findings suggest that students currently show greater interest in cloud technologies and employability-related skills than in specialized software engineering topics.



Among individual workshop sessions, **Codeku – Session 01** achieved the highest attendance with **217 registrations (37.54%)** among the displayed sessions. Similarly, the **Beauty of Cloud** workshop series attracted the highest overall registrations, indicating sustained participant interest in cloud-related learning opportunities.



Participation varied considerably across organizers. **IEEE CS** recorded the highest number of registrations, followed by **ACS**, while **IEEE WIE** and **WIE** organized comparatively smaller events. This indicates that IEEE CS played the most significant role in attracting participants throughout the workshop programme.

The participant demographic analysis demonstrates that the **University of Sri Jayewardenepura** accounted for **689 registrations (63.8%)**, making it the primary contributor to workshop participation. Nevertheless, registrations from numerous other universities indicate successful outreach beyond the host institution.

Faculty-level analysis shows that students from the **Computing** faculty constituted the largest participant group, substantially exceeding registrations from Technology, Engineering, and Applied Sciences. This suggests that technical workshops predominantly attracted computing-related disciplines.

Academic year analysis further indicates that **second-year** and **third-year students** represented the largest participant groups, while fourth-year students participated

comparatively less frequently. This may reflect differences in academic workload, internship commitments, or student availability.

The feedback dashboard demonstrates consistently positive participant perceptions. The average satisfaction score of **8.94**, combined with the predominance of maximum ratings for speaker performance and content relevance, indicates that participants were highly satisfied with workshop quality. However, many respondents did not provide written suggestions, limiting the availability of qualitative insights for future improvements.

6.2 Recommendations

Based on the findings obtained from the Power BI dashboard, several recommendations can be proposed to improve future workshop planning and participant engagement.

Since **Cloud Computing** and **Professional Skills** workshops consistently attract the largest number of participants, future workshop schedules should continue prioritizing these subject areas while expanding their content to meet growing student demand.

Participation from faculties outside Computing remains relatively low. Therefore, organizers should develop targeted promotional campaigns and interdisciplinary workshop topics that appeal to students from Engineering, Applied Sciences, Humanities, Medical Sciences, and Management faculties. Expanding workshop themes may encourage greater participation across diverse academic disciplines.

Although the University of Sri Jayewardenepura contributed the majority of participants, additional collaboration with external universities should be encouraged to increase diversity in institutions. Partnerships with universities that currently have lower participation levels may help expand the reach and impact of future workshops.

Workshop scheduling should consider participant demographics. Since second- and third-year students contribute the largest participant groups, workshops may continue to target these groups while introducing dedicated sessions specifically designed for first-year orientation and final-year career preparation.

The consistently strong performance of **IEEE CS** suggests that its organizational strategies, marketing approaches, and workshop delivery methods could serve as best practices for other

organizing bodies. Therefore sharing successful planning techniques across organizers may help improve participation in future events.

Although participant satisfaction was exceptionally high, the relatively low **20% feedback response rate** indicates an opportunity to strengthen post-event evaluation. Organizers should encourage feedback submission through automated email reminders, QR code surveys distributed immediately after each workshop, mobile-friendly survey forms, or incentives such as certificate release upon survey completion. Increasing the response rate would provide a more representative understanding of participant experiences.

Finally, future Power BI dashboards could be enhanced by incorporating long-term analyses, allowing organizers to monitor participation trends over multiple years, compare workshop performance over time, and evaluate the effectiveness of continuous improvement initiatives.

7. Conclusion

This project successfully achieved its objective of developing an interactive Business Intelligence dashboard using Microsoft Power BI to analyze participation and engagement in technology-related academic workshops conducted by student organizations at the University of Sri Jayewardenepura. The dashboard combined information from various workshop series, allowing the team to access registration data and feedback across all series in a uniform format. The dashboard combined registration data and feedback from multiple workshop series into a common data model and provided meaningful visual insight into the data to enable informed decision-making.

The analysis showed the pattern of participation across workshop categories, faculties, universities, academic years, and organizing bodies. Cloud Computing and Professional Skills workshops attracted the highest levels of participation, while students from the Faculty of Computing and the University of Sri Jayewardenepura represented the largest participant groups. Feedback analysis also indicated high levels of participant satisfaction, speaker effectiveness, and content relevance, demonstrating the overall success of the workshops. At the same time, the relatively low feedback response rate highlighted an area where future workshop organizers can improve participant engagement after events.

The interactive dashboard allows organizers to monitor workshop performance, compare participation trends, evaluate participant demographics, and assess feedback through dynamic visualizations and filtering capabilities. These findings support evidence-based planning, allowing organizers to identify successful workshop strategies, improve promotional activities, and allocate resources more effectively.

Overall, the project has shown how useful Business Intelligence can be in an educational setting, bringing together the data from the various workshops into information that can be used to support action. In the future, the dashboard can be further enhanced by incorporating data from additional academic years, introducing predictive analytics to forecast workshop participation, and integrating real-time data collection to support continuous monitoring and long-term strategic planning.

8. References

- [1] Microsoft, *Microsoft Power BI Documentation*. Available: <https://learn.microsoft.com/power-bi/> (Accessed: Jul. 2026).
- [2] Microsoft, *Power BI Guided Learning*. Available: <https://learn.microsoft.com/training/powerplatform/power-bi/> (Accessed: Jul. 2026).
- [3] R. Kimball and M. Ross, *The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling*, 3rd ed. Hoboken, NJ, USA: John Wiley & Sons, 2013.
- [4] C. Howson, *Successful Business Intelligence: Unlock the Value of BI and Big Data*, 2nd ed. New York, NY, USA: McGraw-Hill Education, 2014.
- [5] W. H. Inmon, *Building the Data Warehouse*, 4th ed. Indianapolis, IN, USA: Wiley Publishing, 2005.
- [6] T. Davenport and J. Harris, *Competing on Analytics: The New Science of Winning*. Boston, MA, USA: Harvard Business Review Press, 2007.
- [7] S. Few, *Information Dashboard Design: Displaying Data for At-a-Glance Monitoring*, 2nd ed. Burlingame, CA, USA: Analytics Press, 2013.
- [8] Microsoft, *Data Modeling in Power BI*. Available: <https://learn.microsoft.com/power-bi/transform-model/desktop-data-modeling> (Accessed: Jul. 2026).
- [9] Microsoft, *Data Analysis Expressions (DAX) Reference*. Available: <https://learn.microsoft.com/dax/> (Accessed: Jul. 2026).
- [10] Microsoft, *Power Query Documentation*. Available: <https://learn.microsoft.com/power-query/> (Accessed: Jul. 2026).